Charpy Transition Temperature Uncertainty and Test-Matrix Design for Reactor Pressure Vessel Steels

An analysis of published Charpy V-notch test data

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This is an analysis of published Charpy V-notch test data. No Charpy tests were performed for this project, no test equipment was operated, and no irradiated material was handled. Every measurement used here was produced and published by the original investigators credited below.

Abstract

A reactor pressure vessel surveillance capsule holds a small, fixed volume of irradiated steel. That volume can be spent on a few full-size Charpy specimens or on many miniature ones. This work asks which choice yields a better-determined ductile-to-brittle transition temperature once both curve-fitting uncertainty and sub-size-to-full-size correction uncertainty are counted. Working from a published compilation of Charpy V-notch records for ferritic reactor pressure vessel steels, it fits transition curves, quantifies their index-temperature uncertainty by resampling, measures the residual disagreement left by a classical specimen-size correction, and propagates both through a Monte Carlo study of complete simulated test programmes. The size-correction residual enters once per programme and therefore does not average away, which places a floor on what a miniature programme can achieve. Under the central assumptions that floor is 11.96 °C, so miniature specimens help only at the smallest material budgets and full-size specimens dominate as the budget rises. The conclusion is conditional: the correction distribution rests on a handful of independently matched material families, no individual miniature geometry is supported on its own, and the result is stated with those conditions attached throughout.

Evidence classes used in this report

Every figure and table below carries one or more of these labels, and its caption says which visual component carries which provenance. The distinction matters because it is the difference between a measurement, an estimate calculated from measurements, and a simulation.

- PUBLISHED SOURCE DATA — observations and values published by the source studies and compiled in the dataset.
- ANALYSIS-DERIVED FROM PUBLISHED DATA — reconciliation and cohort counts, fitted parameters, index temperatures, bootstrap intervals and residuals: quantities this project calculated from published data.
- TRANSFORMED ESTIMATE FROM PUBLISHED DATA — a published smaller-specimen observation mapped into the full-size-equivalent domain by the classical correction. Never a measurement.
- SYNTHETIC — generated by this project from known truth, to validate a method or to simulate a test programme. Not a measurement of anything.

1. Objective and scope

1.1 The research question

A surveillance capsule is irreplaceable. The steel inside it has seen the same neutron exposure as the vessel wall, and when it is gone there is no more of it. The engineer designing the test programme must therefore decide how to spend a fixed volume of material. Full-size specimens give an index temperature that needs no size correction, but a fixed volume yields few of them. Miniature specimens yield many more, at the cost of a correction from the miniature geometry back to full-size behaviour — and that correction carries its own uncertainty.

The question this work answers is which choice gives the better-determined transition temperature once both sources of uncertainty are counted. Four supporting questions follow from it, and each is answered in section 6: how well-determined the index temperature is for a given curve and what property of the sampling drives that (Q1); how that uncertainty depends on specimen count and on where the test temperatures are placed (Q2); how much additional uncertainty the sub-size to full-size correction introduces and whether it differs by product form (Q3); and under what conditions the answer flips (Q4).

In plain terms — why this is a real engineering choice. Charpy specimens are destroyed by the test, and irradiated steel cannot be replaced. A standard specimen is a bar fifty-five millimetres long with a ten-by-ten millimetre cross-section; a miniature one may be a fifth of that in each direction, so the same lump of steel yields tens or hundreds of them instead of a handful. More specimens mean a better-sampled curve. But a small specimen is not a scale model of a large one: it bends and breaks differently, so its transition temperature must be corrected before it can be compared with the full-size number the regulations are written around. This work asks whether the correction costs more than the extra specimens buy.

1.2 What this work is, and is not

This is a test-programme design study. It is not an attempt to predict full-size properties from sub-size measurements; that is a different problem, addressed by a recent machine-learning correlation study discussed in section 7.7. No model is trained here and no prediction is attempted.

In scope: ferritic reactor pressure vessel steels and their welds, absorbed energy against temperature, the index temperature and upper-shelf energy as the extracted properties, the statistical uncertainty in those properties, and the specimen-size effect as it bears on them. Out of scope, and excluded deliberately: austenitic stainless steel, which is face-centred cubic and does not show the ferritic ductile-to-brittle transition the model describes; fracture mechanics and the determination of reference temperatures to the pressure vessel code; and the prediction of embrittlement from composition and fluence.

2. Data source, provenance and license

The primary dataset is the Materials Cloud 2025.30 release, record q4x6v-dyt25, DOI 10.24435/materialscloud:p6-5y. It holds 4,961 Charpy V-notch records compiled from 109 peer-reviewed publications, covering the A533, A508, A302 and 20MnMoNi55 families in base metal and weld, irradiated and unirradiated, at full-size, sub-size and miniature geometries. The earlier Materials Cloud 2024.206 release, record eny5v-www93, is retained as a version-sensitivity dataset. Which release is primary, and why, is a recorded decision taken at the first checkpoint.

Attribution, as the license requires. Lee, I., Merickel, J. W., Kasala Sreenivasulu, Y., Xu, F., Tang, Y., Rittenhouse, J. E., Vakanski, A., and Song, R. Dataset of Charpy V-notch impact test records of nuclear structural materials. Materials Cloud Archive. Licensed CC BY 4.0. The accompanying data descriptor is Lee et al., Comprehensive Toughness Dataset of Nuclear Reactor Structural Materials using Charpy V-Notch Impact Testing, Scientific Data. Both workbooks are vendored in this repository with their archive record snapshots; their SHA-256 digests and the MD5 values published by the archive are verified on every run.

The single most important fact about this dataset is that it is a compilation of 109 separate studies. Different laboratories, machines, calibration practices and reporting conventions are mixed together, and most curves were digitized from figures published in those papers rather than transcribed from tables. Between-study variability is therefore built into any scatter estimate derived from the compilation, and it cannot be separated from real material scatter. That limitation is not a caveat added at the end of this report; it constrains what several of its numbers can mean, and it is carried explicitly wherever they appear.

3. Data cleaning and inclusion criteria

3.1 Reconciliation

Records leave the dataset only for a stated reason, and each is counted once, at the first step that applies. Out-of-scope materials go first, so that no judgement is spent on data that would leave anyway; duplicate curves go last, so that a valid copy is never removed in favour of one that a later step would itself have removed.

Table 1. Cleaning reconciliation. These counts are ANALYSIS-DERIVED: they are the result of this project's exclusion rules applied to the published compilation, not counts any source study reported. Every record that leaves does so at the first step that applies, for a stated reason, and the arithmetic closes. [ANALYSIS-DERIVED FROM PUBLISHED DATA] Source: outputs/wp1/reconciliation_table.csv.

step	criterion	records_removed	records_remaining
Start	Records in the 'Charpy Data' sheet		4961
E1	Out of scope: austenitic stainless steel (SS316 family)	713	4248
E2	Out of scope: ferritic steel outside the brief's named RPV-steel family (15Cr2MoV-A)	38	4210
E3	Not a standard Charpy impact test: striking velocity outside ASTM E23 range (3-6 m/s)	47	4163
E4	Missing essential measurement: test temperature or absorbed energy	0	4163
E5	Missing essential measurement: specimen dimensions (length, width or thickness)	55	4108
E6	Physically impossible value	0	4108
E7	Exact duplicate of an earlier record (all original fields identical)	0	4108
E8	Duplicate curve: same tests already present from another figure or publication	910	3198
End	Records retained after WP1	1763	3198

Of the 4,961 records in the workbook, 1,763 are removed and 3,198 retained. The largest single exclusion is austenitic stainless steel, 713 records, removed because it has no ductile-to-brittle transition of the kind the model describes. Nothing is imputed at any step: a blank cell means "not reported", and a standard notch depth is never assumed for a specimen that does not report one.

3.2 The same tests, reported more than once

The second-largest exclusion, 910 records, is the one that most affects how much confidence the data can support. Several research groups republished the same Charpy curves across conference and journal papers, and the compilers digitized each figure separately. Counting a republication as a new experiment would overstate the amount of independent evidence and understate scatter. Copies are identified by one-to-one point matching within a tolerance scaled to the plotted energy axis, and the copy kept is chosen by a hierarchy fixed before any curve was fitted: tabulated values over digitizations, then the original source over a republication, then fewer unresolved labelling problems, then a deterministic tie-break independent of the data. Two series matched only after one constant temperature offset — the signature of two digitizations of one figure with differently calibrated axes; that test runs only on publication pairs with documented shared provenance, under recorded safeguards. An alternative keep rule is carried as a sensitivity cohort, so the influence of the choice is measured rather than assumed.

3.3 What defines one curve, and which curves are fittable

Records form one transition curve only if they come from one publication and one source figure and agree on material identity and heat chemistry, specimen geometry, irradiation condition, material condition and test-condition statement. A curve never spans two publications, because between-laboratory differences would then enter what is read as a single curve's scatter. Applying the population rules — removing records whose condition labels the source contradicts, records whose irradiation status could not be resolved, and two secondary material groups — leaves a primary population of 2,734 records in 191 curves from 47 publications.

A pre-fit screening rule then selects the curves sufficiently sampled to attempt a fit: a minimum specimen count, at least two distinct temperatures inside the transition band, enough upper-band points to show the rise has stopped, and a coldest point low enough relative to the curve's own upper reference. Passing means only that a fit may be attempted, not that the model will fit, that it will be well conditioned, or that its parameters will be reasonable — and a curve whose fit later fails is reported, never quietly dropped. The rule yields the primary cohort of 170 curves and 2,588 records; a deliberately looser rule would admit 182 curves and a deliberately stricter one, matching historical surveillance-programme specimen counts, only 60.

Table 2. The curve cohorts frozen at checkpoint CP2, ANALYSIS-DERIVED from the published records by this project's grouping and screening rules. Each differs from the primary cohort in exactly one judgement, so any later change in a result is attributable to one decision. Membership is fingerprinted and re-checked on every run. [ANALYSIS-DERIVED FROM PUBLISHED DATA] Source: outputs/wp2/cohort_manifest.csv.

cohort	role	curves	records
primary	primary analysis	170	2588
looser	inclusion-rule sensitivity	182	2692
stricter	inclusion-rule sensitivity	60	1333
alt_keep_rule	duplicate keep-rule sensitivity	170	2628
plus_unresolved_irradiation	population sensitivity	179	2704
plus_additively_manufactured	population sensitivity (secondary material)	175	2654
piping_secondary	secondary group	15	181

All seven cohorts were frozen at the second checkpoint, fingerprinted, and are re-checked on every run; the pipeline stops if any membership drifts. Later stages read them rather than regenerating them, so no downstream result can be improved by quietly adjusting an inclusion threshold.

4. Methods

4.1 The transition-curve model and its lower shelf

Absorbed energy against temperature is described by the conventional hyperbolic tangent form, in which energy rises from a lower shelf to an upper shelf through a mid-transition temperature with a characteristic half-width. The model is empirical. It has no physical derivation, it is symmetric where real transitions often are not, and it cannot represent a falling upper shelf; where it misfits, the misfit is diagnosed rather than removed.

The lower shelf is fixed at 2.7 J for full-size specimens, the value used in the Oak Ridge sub-size Charpy work, and fitted for miniature geometries, for which no single defensible fixed value exists across the nine geometries present. The split rests on the literature, the estimand and synthetic evidence that predated the choice, not on how many published curves it made succeed. Fitting the shelf everywhere is regenerated on every run as a named sensitivity, and moves the full-size index temperature by a median of well under a degree.

4.2 Upper shelf, index temperature, and why miniature curves get none

Upper-shelf energy is the fitted upper asymptote, reported together with a diagnostic saying whether any specimen was actually tested high enough to support it. The index temperature is the temperature at which the fitted curve crosses 41 J, solved analytically. It is computed for full-size V-notch curves only, and only when the crossing lies inside the tested temperatures; an extrapolated crossing is not reported as a value.

Two support diagnostics travel with every index temperature, and neither withholds it: one says where 41 J sits on the fitted rise, exposing a crossing that rests on a nearly flat part of the curve; the other says whether the measured energies bracket 41 J, exposing a crossing supplied mainly by the model. They answer different questions and are reported separately, because a curve can pass one and fail the other.

No index temperature is computed for miniature curves. The applicable standard states that miniaturized transition data are not directly comparable with full-size data and that correlation procedures must be employed, and the published procedures need inputs this dataset lacks. Multiplying 41 J by a volume ratio is supported by no source read here, so no such criterion was invented; comparing the two sizes is the job of the correction in section 4.4, with its uncertainty attached.

4.3 Per-curve uncertainty

Uncertainty is obtained by resampling. The primary method is a nonparametric pairs bootstrap: each replicate draws complete specimen records — temperature and energy together — with replacement and refits them with the unchanged procedure. It assumes no scatter model, does not assume the model is correct, and exposes how fragile an estimate is to which test temperatures happen to be present, which is the question at issue. A fixed-design wild bootstrap, holding the temperatures and perturbing only residuals, runs on every curve as a sensitivity and is never promoted, because validation showed it anti-conservative.

Intervals are nominal 95 % percentile bootstrap intervals. They were not calibrated to exact coverage, and section 5 reports what coverage they actually achieved on synthetic data. A formal interval is reported only when the target is finite in at least 95 % of the replicates requested; failed replicates stay in the denominator. That rule was fixed before any interval existed and was not relaxed afterwards.

4.4 The classical size normalization

Smaller-specimen curves are mapped into the full-size-equivalent domain by two published relations: energy scaled by the ratio of nominal fracture volumes, the ligament-volume convention the dataset's descriptor writes as $B(W - A)^2$, and temperature shifted by a logarithmic function of that same volume, from the Oak Ridge correlation report developed on this material family. Both depend only on the smaller specimen's own published dimensions and fixed published constants, so no full-size answer can enter its own correction — checked numerically on every run. The transformed points are refitted with the unchanged fitter under the full-size convention, and the residual is the difference from the measured full-size index temperature, signed so that positive means the correction places the transition warmer, the conservative direction.

Several published procedures could not be run, and none of their inputs was invented. The procedure the correlation report itself recommends weights a brittle and a ductile factor by percent shear, and this dataset has no shear-fracture column for any record. Two others need a stress-concentration factor tabulated for only a few geometries, and two more need exponents calibrated against the very full-size values a correction would then be scored on, which is circular.

4.5 The Monte Carlo test-programme simulator

The design question is answered by simulating complete test programmes against a known true curve. One replicate picks a true curve, a geometry and a material budget; derives the specimen count; places temperatures by a declared strategy using only prior information a designer would have; draws energies from a scatter model calibrated to the published residuals; for a smaller geometry applies the classical correction back into the full-size domain; fits with the unchanged procedure; and records the error in the recovered index temperature.

Three features matter most. Test temperatures live in the real world: snapped to a five-degree grid and clipped to a physical envelope whose cold end is liquid nitrogen, in raw specimen temperature rather than the corrected frame — so a geometry with a large temperature shift cannot reach a cold transition, and that shows up as infeasibility, not as a poor number. The correction residual is drawn once per programme, not per specimen, because the quantity of section 4.4 is defined per matched curve; a per-specimen draw would average away and imply, absurdly, that testing enough small specimens removes a systematic correlation error. The estimator is the one this project reports, including its refusals: a programme whose crossing falls outside its own tested range has no index temperature and is counted unavailable.

Performance is the root-mean-square error of the recovered index temperature; a cell is unusable if the estimator produced a value in fewer than 95 % of replicates. A winner needs a difference exceeding both a practical tolerance and three times its paired Monte Carlo standard error; otherwise the comparison is a tie.

In plain terms — what a bootstrap does, and why the correction is drawn once per programme. A bootstrap answers "how much would this answer have moved if the experiment had come out slightly differently?" by building many pretend experiments out of the data actually collected, refitting each one, and looking at the spread of answers. It measures scatter and sampling luck. It cannot measure an error shared by every specimen in the programme. The size correction is exactly such an error: one formula, applied to one curve, either fits that material or does not. Drawing it once per programme keeps that distinction honest, and it is why testing more miniature specimens narrows the curve fit without moving the correction error at all.

5. Fitter validation

Nothing was trusted on real data before it was shown to recover known truth on synthetic data. All results in this section are SYNTHETIC: curves generated from known parameters, with controlled scatter, at a fixed seed.

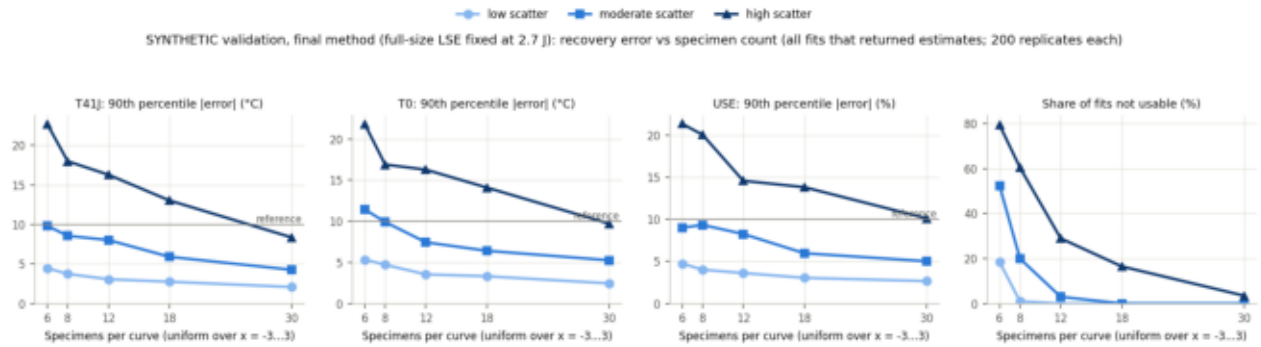


Figure 1. Fitter validation against known truth: index-temperature error against the number of specimens, at three scatter levels. Curves generated from known parameters; no published measurement appears in this figure. [SYNTHETIC] Source: outputs/wp3/synthetic_error_vs_specimens.png.

The fitter recovers its parameters well where the sampling identifies them and badly where it does not, and the failures are reported rather than tuned away. Three limits matter and are stated plainly. Where every specimen lies below the true transition, a small fraction of fits still pass every diagnostic while the upper-shelf energy is wrong by most of its value — the diagnostics are computed on the fitted curve, so a confidently wrong fit can look supported. Where scatter is high, many specimens do not rescue the fit. And near a threshold the diagnostics are conservative, declining fits whose index temperature is in fact accurate.

The resampling procedure was validated the same way. On data lying exactly on the model it reproduces the original fit; its results do not depend on the number of worker processes; and every replicate is accounted for, including those that produce no value. Coverage was measured, not assumed, and it is not exactly the nominal level: the primary pairs intervals covered the truth in about 0.90 of well-sampled synthetic cases, 0.93 at high scatter and 0.97 for a weakly sampled design, with a Monte Carlo standard error of about 0.015. The fixed-design alternative under-covered everywhere, reaching only 0.39 in the weak design, which is why it is reported as a sensitivity and never as the primary result.

The simulator was validated in turn: 20 validation rows, 20 of them passing, covering exact reproduction of the real fits by the simulator's own analysis path, the absence of any route by which the true answer could leak into the design, the physical reachability of the requested temperatures, and the requirement that the temperature shift be applied exactly once. A further 11 independent checks recompute the published results from the raw geometry and the declared rules without importing the code they check, and 9 do the same for the size normalization.

6. Results

6.1 Fitting the published curves (Q1)

All 170 curves of the primary cohort were fitted: 134 full-size and 36 miniature. Each receives exactly one outcome, and the categories are exhaustive.

Table 3. Fit outcomes for every curve of the primary cohort, mutually exclusive and exhaustive, ANALYSIS-DERIVED from the published measurements. No curve is silently dropped: a curve the fitter cannot resolve is reported as such. [ANALYSIS-DERIVED FROM PUBLISHED DATA] Source: outputs/wp3/fit_outcomes.csv.

outcome	curves	full-size_curves	miniature_curves
optimizer failure	0	0	0
non-physical (inadmissible)	0	0	0
boundary-limited	18	1	17
weakly identified	35	29	6
usable fit, index temperature unavailable	13	0	13
usable fit, index temperature calculated	104	104	0
TOTAL	170	134	36

There were 0 optimizer failures and 0 inadmissible solutions. 18 fits rest on a bound and 35 converged but are not identified by their sampling; these are reported as such rather than presented as results. 104 curves yield a usable fit with an index temperature, and 13 yield a usable fit with none, all miniature, because no index criterion exists for those geometries. The answer to Q1 begins here: whether an index temperature is well determined is decided less by how many specimens were tested than by how many of them landed in the transition rather than on a shelf.

6.2 Per-curve uncertainty and its cross-check (Q1, Q2)

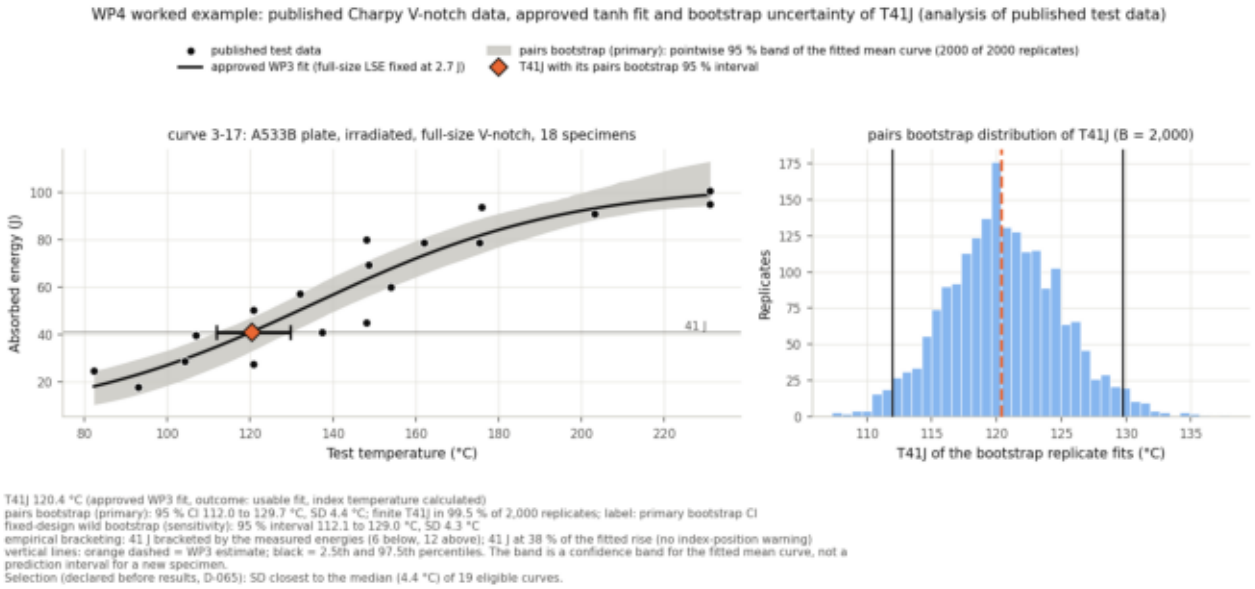


Figure 2. A worked example of one fitted transition curve. The plotted energies are PUBLISHED SOURCE DATA; the fitted curve, the pointwise bootstrap band of the fitted mean curve and the index temperature with its interval are ANALYSIS-DERIVED FROM PUBLISHED DATA. The band is a confidence band for the mean curve, not a prediction interval for a new specimen. Curve 3-17, A533B plate, irradiated. [PUBLISHED SOURCE DATA + ANALYSIS-DERIVED FROM PUBLISHED DATA] Source: outputs/wp4/worked_example_t41j_bootstrap.png.

Across the full-size curves with a usable fit and a reportable primary interval, the bootstrap standard deviation of the index temperature has a median of 4.9 °C, with a tenth-to-ninetieth percentile range of 2.9 to 9.1 °C, and a median nominal interval width of 19.4 °C.

Table 4. Which curves received which kind of interval, ANALYSIS-DERIVED by resampling the published measurements. Intervals are nominal percentile bootstrap intervals, not intervals calibrated to exact coverage. The unavailable intervals are a result about test-matrix fragility, not a gap in the table. [ANALYSIS-DERIVED FROM PUBLISHED DATA] Source: outputs/wp4/interval_curve_accounting.csv.

method	geometry_category	target	curves	primary_bootstrap_ci	diagnostic_bootstrap_interval	interval_unavailable
pairs	full-size	t41j	134	66	14	54
pairs	full-size	use	134	104	30	0
pairs	full-size	t0	134	104	30	0
pairs	miniature	t0	36	13	23	0
pairs	miniature	use	36	13	23	0
wild	full-size	t41j	134	97	29	8
wild	full-size	use	134	104	30	0
wild	full-size	t0	134	104	30	0
wild	miniature	t0	36	13	23	0
wild	miniature	use	36	13	23	0

The accounting itself carries a finding. Of the 134 full-size curves, 66 receive a primary interval, 14 a diagnostic-only interval, and 54 none: in those, enough resampled test matrices lose the cold specimens that place the crossing that no nominal interval can honestly be quoted. Holding the temperatures fixed leaves only 8 unavailable. That gap is part of the answer to Q2: a test matrix whose index temperature rests on one or two cold specimens is fragile, and that fragility is invisible to any method conditioning on the temperatures actually tested.

Against published work the magnitudes are sane and honestly larger. Over the 66 curves with a primary interval and a median of 14 specimens, the median bootstrap standard deviation is 4.87 °C against 3.52 °C for a designed eighteen-specimen matrix on one well-characterized forging — a ratio of about 1.38 — and it falls with specimen count close to the inverse square root, slope -0.32. Being larger than a single-material, single-laboratory study is the expected result for curves drawn from many studies with uneven temperature layouts, and is reported as such rather than as agreement.

6.3 The specimen size effect (Q3)

Of the candidate pairs enumerated across the cohort, 18 are strictly matched geometry comparisons, drawn from 10 independent material-condition families in 5 publications. 10 comparisons from 7 families in 5 publications reach the pre-registered primary analysis set. No cross-study comparison survives: without a heat identifier, the reported composition is the only heat proxy, and it never matches exactly across publications.

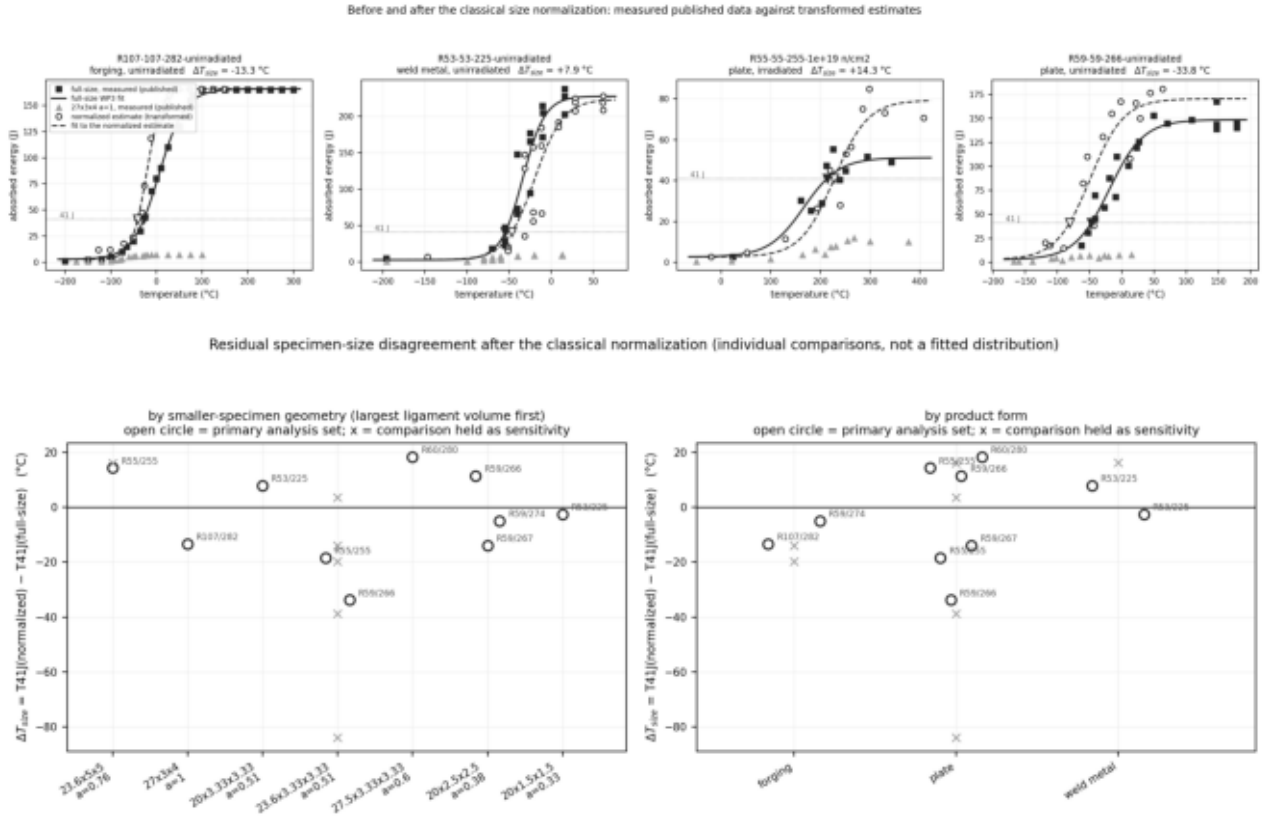


Figure 3. The specimen-size normalization and what it leaves behind. (a) Curves before and after the classical size normalization, for four matched material-condition families. (b) The residual disagreement that survives normalization, every comparison shown individually, by smaller-specimen geometry and by product form. In panel (a) the full-size and smaller-specimen points are PUBLISHED SOURCE DATA, the mapped points are a TRANSFORMED ESTIMATE FROM PUBLISHED DATA, and both fitted curves and the refit of the mapped points are ANALYSIS-DERIVED FROM PUBLISHED DATA. In panel (b) each plotted residual is ANALYSIS-DERIVED: the difference between a transformed estimate and a fitted full-size index temperature. Individual points are shown rather than a fitted distribution, because there are too few independent families to justify smoothing. [PUBLISHED SOURCE DATA + ANALYSIS-DERIVED FROM PUBLISHED DATA + TRANSFORMED ESTIMATE FROM PUBLISHED DATA] Source: outputs/wp5/before_after_normalization.png; outputs/wp5/size_correction_residuals.png.

The normalization helps substantially: the median absolute disagreement in mid-transition temperature falls from 66.9 to 16.2 °C, and 8 of the 10 primary comparisons improve on that measure, 9 on upper-shelf energy. It does not, however, collapse the curves.

Table 5. The residual index-temperature disagreement left by the classical size normalization. Each residual is ANALYSIS-DERIVED from a TRANSFORMED ESTIMATE: the difference between a mapped smaller-specimen index temperature and the matched full-size one. Row labels are shortened for the page: PRIMARY is the pre-registered primary analysis set, and ALL STRICT is every strictly matched comparison with a residual, that is the primary set together with those held as sensitivity; family-weighted means each family contributes the mean of its own comparisons and then counts once. The family is the unit of independence, so the family-weighted rows are the ones carried forward. [ANALYSIS-DERIVED FROM PUBLISHED DATA + TRANSFORMED ESTIMATE FROM PUBLISHED DATA] Source: outputs/wp5/size_correction_summary_overall.csv.

subset	weighting	n	n_families	n_studies	mean_c	sd_c	min_c	max_c
PRIMARY	comparison-weighted	10	7	5	-3.494	16.66	-33.82	18.32
PRIMARY	family-weighted	7	7	5	-3.494	11.43	-14.01	18.32
ALL STRICT	comparison-weighted	17	10	5	-9.141	26.05	-83.88	18.32
ALL STRICT	family-weighted	10	10	5	-10.8	21.09	-48.94	18.32

The residual index-temperature disagreement has a family-weighted mean of -3.49 °C and a family-weighted standard deviation of 11.43 °C, with individual comparisons spanning -33.8 to 18.3 °C; a robust scale computed from the median absolute deviation gives 11.34 °C, so the spread is not an artefact of one extreme comparison. Weighting by family rather than by comparison matters, because several geometries measured on one plate are not independent observations of the size effect. The more conservative pool, using every strictly matched comparison rather than only those meeting the primary fit-quality requirement, gives -10.80 °C and 21.09 °C from 10 families; it is carried forward as a required sensitivity.

Q3's second half is answered negatively, and that is a result. No individual miniature geometry reaches the pre-registered threshold of three independent families from two publications — a threshold not lowered after nothing met it. Among product forms only plate qualifies (4 families, 3 publications; family-weighted mean -2.21 °C, standard deviation 14.62 °C); forging has 2 families and weld metal 1, so no standalone estimate is made for either. By irradiation state only unirradiated qualifies (5 families; -1.69 °C, 12.80 °C); irradiated has 2, so no separate irradiated correction is estimated, and it is not assumed that irradiation and geometry fail to interact.

The two comparisons that the normalization makes worse are both from one severely embrittled, low-toughness plate, where the miniature specimens transitioned at or above the full-size temperature — the opposite of the direction the published shift assumes. Adding a positive shift to data whose measured shift runs the other way necessarily worsens the agreement. This is reported, not removed.

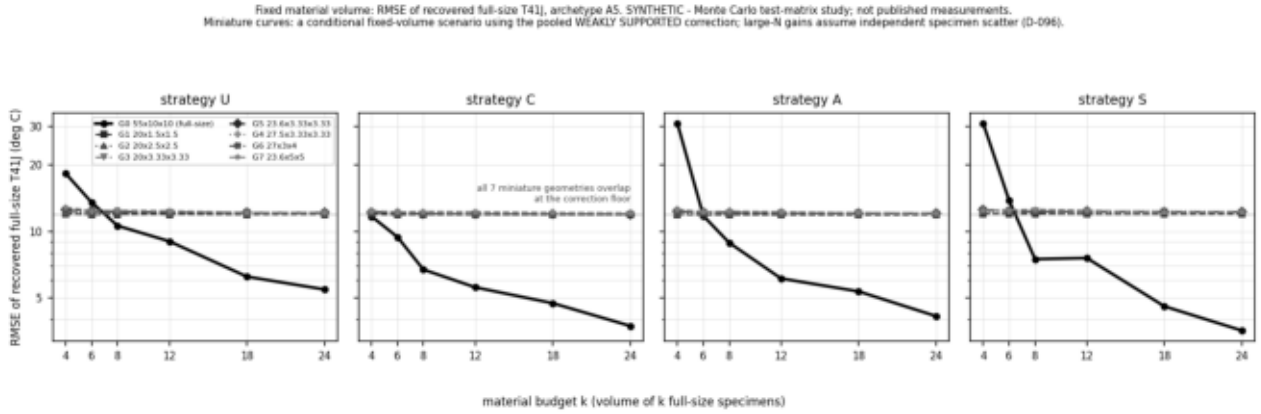
Finally, the choice of published correlation moves the answer more than the scatter in it: across the pre-declared alternatives the family-weighted bias runs from -3.5 to -23.2 °C while the family-weighted standard deviation stays between 11.1 and 11.9 °C. The scatter about a chosen correlation is stable; the location of the correction is not.

6.4 The test-matrix study (Q2, Q4)

> In plain terms — the answer, before the numbers. Fitting a curve gets more accurate as specimens are added, roughly as one over the square root of the count. Correcting a miniature curve to full-size behaviour does not: that error is set by how well the correlation suits the material, and testing more specimens does not improve it. So a miniature programme's total error falls with specimen count only until it reaches the correction error, and then stops. A full-size programme has no such floor. The question is simply whether the budget is small enough that the extra miniature specimens still matter more than the floor.

What is compared, and what could not be. Full-size is simulated, and so are the seven smaller MCVN/miniaturized geometry scenarios the size analysis supports. Conventional ASTM E23-type sub-size specimens are not represented at all: the dataset's only two such curves have four points each and fail every screening rule, so no correction for that class was ever estimated, and none was invented to fill the gap. The brief asks for full-size against sub-size against miniature; this dataset can answer only two of those three, and the third is reported as a negative result rather than approximated by the miniature arms.

The headline comparison holds material volume fixed across 9 true curves, 7 miniature geometries, six budgets and four placement strategies — 1,512 paired comparisons. The outcome is 1,038 full-size wins, 234 miniature wins, 240 practical ties and 0 ineligible. Every one of these is a conditional fixed-volume scenario using the pooled, WEAKLY SUPPORTED correction.



Conventional E23 sub-size: not estimated — insufficient included evidence.

Figure 4. The fixed-material-volume comparison, one panel per placement strategy. Full-size is simulated, and so are the seven approved smaller MCVN/miniaturized geometry scenarios. Conventional ASTM E23-type sub-size performance is NOT ESTIMABLE from this dataset under the approved rules, and no external correction was invented to fill that gap, so no sub-size arm appears here. Every miniature curve is a conditional fixed-volume scenario using the pooled, WEAKLY SUPPORTED correction, not a geometry-specific claim. Simulated programmes throughout. [SYNTHETIC] Source: outputs/wp6/fixed_volume_comparison.png.

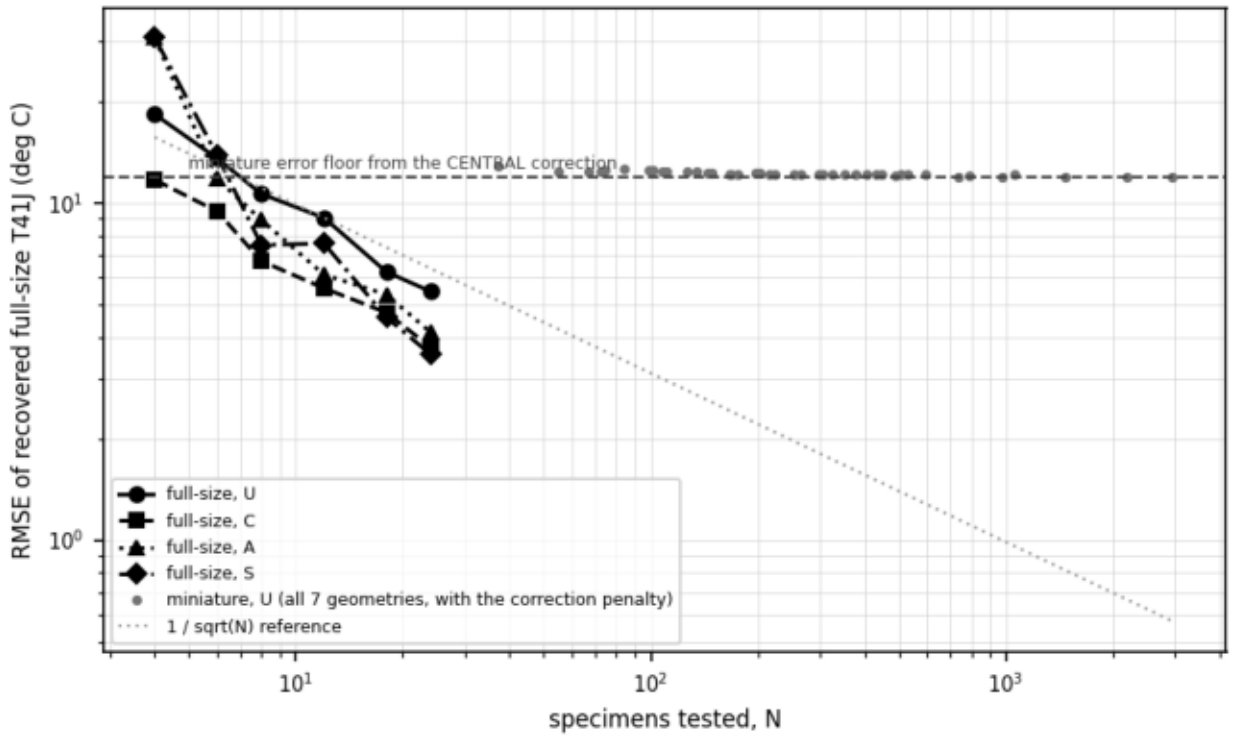


Figure 5. Index-temperature uncertainty against specimen count for the median true curve (archetype A5), one series per test-temperature placement strategy, with the square-root-of-N reference. The horizontal line is the error floor implied by the central size correction: a full-size programme keeps improving with specimen count, a miniature programme stops at the floor. Simulated programmes throughout. [SYNTHETIC] Source: outputs/wp6/rmse_vs_specimen_count.png. The top 44 pixels of the source image, which contain only a title clipped in the source file itself, are masked at the report layer; the approved file is unchanged and no plot content is removed.

The mechanism is visible in both figures: every miniature geometry converges to the same floor, because under a pooled correction what limits them is the correction, not how many specimens each geometry fits. That floor is 11.96 °C, the root-sum-square of the correction bias and its residual standard deviation, and it does not move with specimen count. Under the conservative correction it is 23.69 °C.

Stated in the conditional form these results support: at the smallest budget miniature specimens win for the uniform, anchored and model-based adaptive placements but not the clustered one; around the next budget the answer is placement-dependent and often tied, with miniature wins up to a budget of 6 full-size equivalents; from 8 they win no comparison under any strategy, true curve or geometry; and from 18 full-size wins every one. This is a conditional fixed-volume scenario using the pooled, WEAKLY SUPPORTED correction; it is not a geometry-specific claim, and not a statement that full-size specimens are preferable in general.

How well would the correction have to be known? The threshold is the correction residual standard deviation at which a miniature programme would match a full-size one.

Table 6. How well the size correction would have to be known for a miniature programme to match a full-size one, for the median true curve at one geometry. Simulated; compare with the analysis-derived correction standard deviation reported in the text. [SYNTHETIC] Source: outputs/wp6/correction_threshold.csv.

strategy	k_budget	n_mini	rmse_full_c	sd_star_simulated_c	sd_star_semi_analytic_c
A	4	84	30.86	31.08	30.54
A	8	168	8.883	7.773	7.841
C	4	84	11.69	10.79	10.87
C	8	168	6.713	5.329	5.337
S	4	84	30.86	30.71	30.48
S	8	168	7.489	5.943	5.931
U	4	84	18.28	17.55	17.63
U	8	168	10.61	9.72	9.645

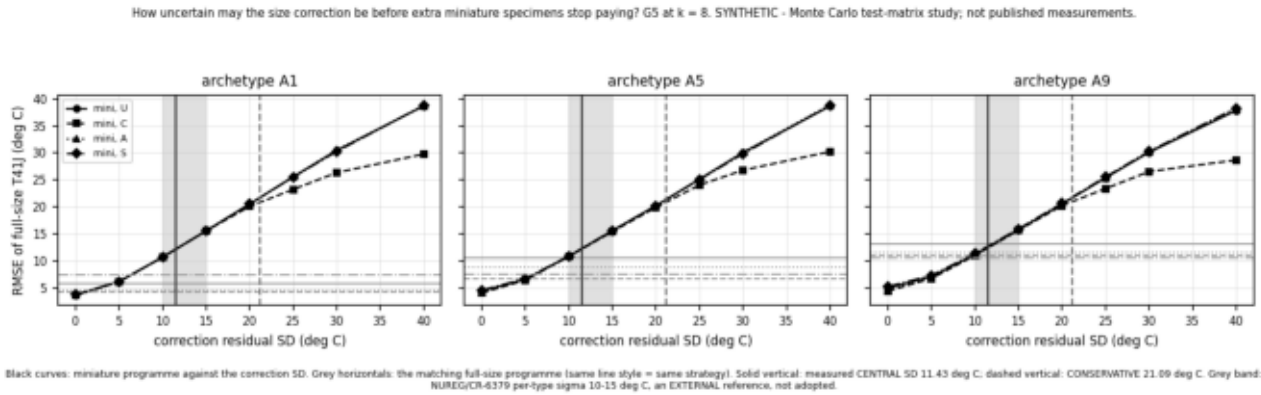


Figure 6. How uncertain the size correction may be before extra miniature specimens stop paying. Miniature programme error against the correction residual standard deviation, against the matching full-size programme, with the analysis-derived central and conservative values marked. Simulated programmes. [SYNTHETIC] Source: outputs/wp6/correction_uncertainty_flip.png.

For the median true curve at a budget of eight full-size equivalents, that threshold lies between 5.3 and 9.7 °C depending on the placement strategy — below the 11.43 °C estimated from the matched published curves. Across all three true curves examined it spans 1.3 to 12.3 °C. At the smallest budget the threshold is strongly strategy-dependent: 31.08, 30.71 and 17.55 °C for the anchored, model-based adaptive and uniform designs, but only 10.79 °C for the clustered one. The 11.43 °C analysis-derived correction standard deviation lies below the first three and above the clustered threshold, which is exactly the observed winner pattern: in this conditional fixed-volume scenario miniature specimens win at this budget under those three placements and not under the clustered one. Across all three true curves these thresholds span 6.2 to 36.9 °C. That is the mechanism behind the flip: the correction uncertainty sits below the relevant threshold at the smallest budget and above it thereafter. The simulated and semi-analytic thresholds agree closely, which confirms the floor arithmetic rather than merely restating it.

Placement and prior knowledge (Q2, Q4).

For full-size programmes placement matters a great deal. A tightly clustered design is best when the transition is known to within a few tens of degrees and collapses when it is not, failing the availability gate rather than merely losing accuracy — it misses the transition altogether. A wide uniform design is robust and never best. A simple up-and-down staircase, spending every specimen near the criterion energy instead of on shelf anchors, outperforms the batched model-based adaptive design by a wide margin. That last finding is itself a caution: the adaptive conclusion depends on which adaptive algorithm is used, so no general claim about adaptive designs is made here. For miniature programmes placement barely matters, because the correction floor dominates whatever the design.

Physical reachability, which no penalty assumption can change. The coldest full-size-equivalent temperature a geometry can reach is liquid nitrogen plus that geometry's own temperature shift. Of the 104 real fitted curves used as truths, 22 have transitions the smallest geometry cannot reach and 2 are beyond an intermediate one. The limit bites before the hard boundary: curves reachable but close to it lose the sub-criterion specimens from too many programmes to be usable. For a cold-transition material the question is not which size is more precise but whether the smaller one can be tested at all, so feasibility failures are reported separately, never averaged into a performance number.

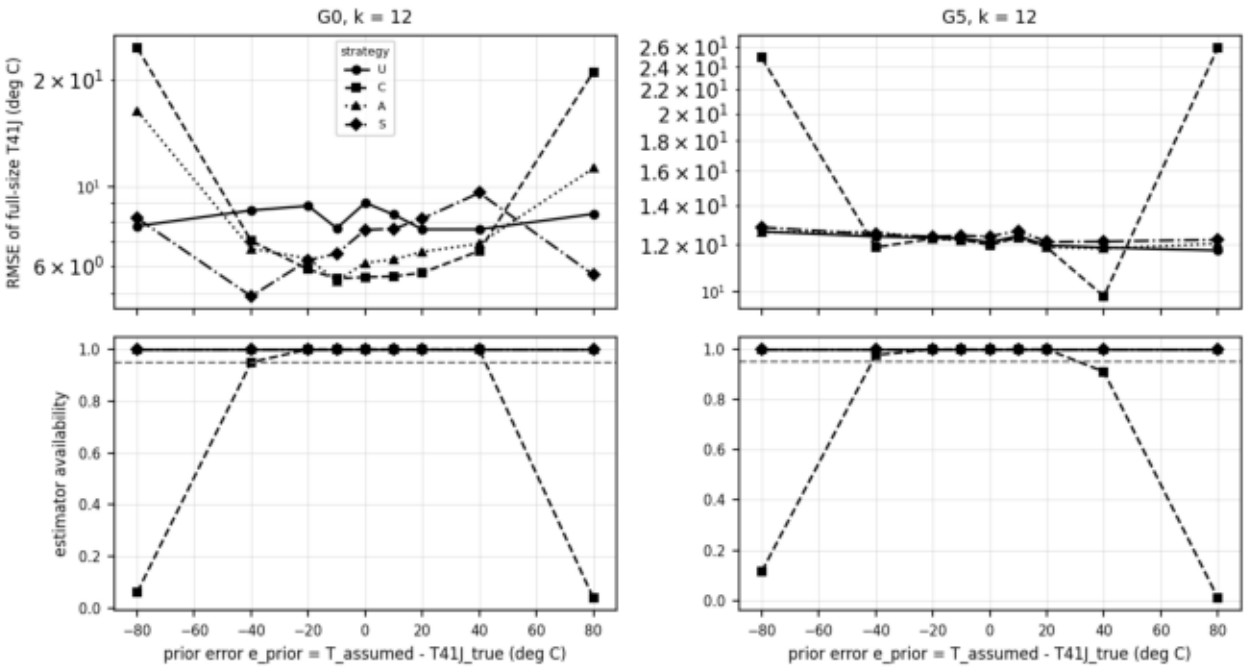


Figure 7. Prior knowledge of the transition location. Error and estimator availability against the error in the assumed transition temperature, with the availability gate marked. Where a design misses the transition it is declared unusable rather than merely imprecise. Simulated programmes. [SYNTHETIC] Source: outputs/wp6/prior_error_vs_performance.png.

6.5 Sensitivities

Table 7. The pre-registered sensitivity variants, all simulated. A positive median difference favours miniature specimens. The size-correction assumptions dominate; the scatter model does not. [SYNTHETIC] Source: outputs/wp6/sensitivity_summary.csv.

variant	median_delta_rmse_c	miniature_wins	full_size_wins	tied	ineligible	changes_conclusion
V1 scatter C1-robust	-3.67	0	33	15	0	False
V10 normalization S3 temperature Lucon	-3.489	0	34	14	0	False
V11 normalization S5 temperature Wallin	-11.28	0	48	0	0	False
V12 normalization S4 temperature Towers	-16.14	0	48	0	0	False
V13 normalization S6 KLST as recorded	-2.881	2	30	16	0	False
V14 specimen-level correction draw	3.357	34	0	14	0	True
V15 true LSE 6 J, analysed at 2.7 J	-3.731	6	30	12	0	False
V16 T41J_true = -150 deg C	-4.679	0	0	0	36	False
V17 T41J_true = 100 deg C	-4.155	0	30	6	0	False
V18 machining allowance c = 0.5 mm	0.5533	20	16	12	0	True
V19 machining allowance c = 1.0 mm	-0.38	18	16	14	0	True
V2 scatter C2	-4.591	0	36	12	0	False
V20 energy-scale injection (optional)	-2.242	3	26	19	0	False
V21 WP2 looser cohort	-2.712	4	30	14	0	False
V22 WP2 stricter cohort	-3.178	6	29	13	0	False
V3 scatter C3-centred	-2.989	3	26	19	0	False
V4 scatter C3-uncentered	-2.671	4	28	16	0	False
V5 miniature-inflated scatter	-2.728	0	27	21	0	False
V6 correction CONSERVATIVE	-13.59	0	48	0	0	False
V7 correction empirical 7-family	-2.001	5	24	19	0	False
V8 correction empirical 10-family	-13.31	0	48	0	0	False
V9 normalization S1 energy (Bb)^3/2	-2.96	1	30	17	0	False

Of the 22 pre-registered variants, 3 change the conclusion, and which ones do is informative. The size-correction assumptions dominate. Several correction-side sensitivities strengthen the full-size result: the conservative residual pool, the ten-family empirical pool, and two alternative published temperature correlations each produce full-size wins in every comparison run. Each remains a conditional fixed-volume scenario and alters only the stated correction assumption relative to the central case. The scatter model matters much less, shifting the median difference by at most about a degree and never reversing its sign — which is worth stating plainly, because the scatter model was identified in advance as the weak link and turns out not to be the one that decides the answer. The programme-level correction draw is decisive: redrawing the residual per specimen, carried as a labelled illustration precisely to show this, lets the correction error average away and reverses the conclusion.

Two machining-allowance variants favour miniature specimens. A machining allowance applied to every geometry costs the smaller specimens proportionally more specimens, yet improves their standing, because it cuts the full-size count into the range where full-size error is steep in specimen count while the miniature error already sits at its floor. The idealized volume accounting used throughout is therefore, if anything, favourable to full-size; it does not flatter the smaller specimens.

Product form was never tested in the simulation. Substituting the plate stratum's own bias and residual scale into the floor arithmetic gives 14.78 °C rather than 11.96 °C. The plate-only arithmetic gives a larger correction-error floor than the pooled central case. No simulated winner or tie labels were recomputed for this plate-only scenario, because it was never part of the frozen experiment grid. A larger floor would be expected to reduce rather than improve the miniature advantage, but that is a directional inference, not a simulated result.

Interval coverage: a practical warning.

Table 8. Interval coverage on simulated programmes. Coverage A is of the programme's own corrected estimand, which is what the bootstrap samples; coverage B is of the true full-size index temperature. For full-size programmes they coincide; for miniature ones they must not. [SYNTHETIC] Source: outputs/wp6/interval_coverage.csv.

geom_code	k_budget	n	coverage_A_own_estimand	coverage_B_true_t41j	median_width_c
G0	6	6	0.9611	0.9611	85.66
G0	12	12	0.925	0.925	34.61
G0	24	24	0.895	0.895	19.82
G5	6	126	0.915	0.265	8.882
G5	12	252	0.965	0.175	6.265
G5	24	504	0.935	0.105	4.429

A bootstrap interval on a miniature-derived index temperature covers the programme's own corrected estimand well, reaching 0.94 at the largest count tested, but covers the true full-size index temperature only 0.27, 0.17 and 0.10 of the time at 126, 252 and 504 specimens: the interval narrows as specimens are added while the correction residual does not, so the quoted uncertainty grows steadily more misleading. This is a mismatch between sampling uncertainty and full-size-equivalent uncertainty after an uncertain correction, not a failure of the resampling method.

7. Uncertainty and limitations

This section is the most important in the report. Nothing in section 6 should be read without it.

7.1 What the data are

The compilation draws on 109 source publications with different machines, calibrations and reporting conventions, and most of its curves were digitized from published figures. Between-study variance is therefore present in every scatter estimate here and cannot be separated from real material scatter. The dataset contains no heat or lot identifier, so reported composition is the only heat proxy — which is what empties the cross-study matched class and confines the size analysis to five publications. It contains no shear-fracture-appearance and no lateral-expansion data at all, which is why the fracture-mode-weighted procedure the correlation report itself recommends could not be run; nothing was imputed to run it. Exact notch dimensions are unreported for the majority of full-size curves; they are treated as full-size V-notch data and flagged, and are never described as confirmed standard geometry.

Several sources were read only in part, and the report does not pretend otherwise: one key fitting paper was available only as an abstract, and the abstract does not contain the statements attributed to it earlier in this project's history; one uncertainty paper was read as a non-peer-reviewed preprint; and a lower-shelf convention attributed to surveillance-report practice could not be retrieved and is therefore not used. The irradiation status of four references could not be resolved from accessible full texts, and those records sit outside the primary population rather than being guessed.

7.2 What the model is

The hyperbolic tangent form is empirical and has no physical derivation. Charpy absorbed energy is an empirical index, not a fracture-mechanics material property, and the index temperature is a correlation parameter used by regulation, not a material constant. The intervals reported here are nominal, not calibrated, and their measured synthetic coverage is given in section 5 rather than assumed. The scatter pool used to calibrate the simulator is empirical published-data residual scatter and is never treated as pure material scatter: it mixes specimen-to-specimen variation, laboratory and machine differences, calibration and reporting differences, digitization error and model-shape mismatch.

7.3 What the correction is

The correction distribution rests on 7 independent matched families from 5 studies. Every summary statistic behind it is a single-digit sample. No individual miniature geometry is directly supported, so no geometry-specific correction or performance claim is made anywhere in this report; the pooled distribution assumes one correction across geometries whose ligament volumes differ by more than an order of magnitude, an assumption these data neither establish nor refute. The evidence is entirely from miniaturized specimens, which the applicable standard distinguishes explicitly from conventional reduced-thickness sub-size specimens; the dataset contains no usable curve of the latter kind, so nothing here is evidence about them.

A scalar volume ratio cannot remove the size effect, for reasons that are physical rather than statistical. Thickness changes the constraint at the notch root: a thick specimen is closer to plane strain, where the stress state is strongly three-dimensional and favours cleavage, while a thin one can deform out of plane and behaves more ductilely at the same temperature. That is a temperature effect, not an energy effect, which is why a second and separate temperature correction is needed at all. A larger specimen also samples more material and is more likely to contain the worst defect, pushing it towards brittle behaviour. And the components of absorbed energy — initiation, propagation, and the shear lips at the specimen sides — do not scale together. The miniature specimens here are not photographic reductions of a full-size specimen: their notch angles, root radii and relative notch depths differ from the standard, and a sharper notch root is known to raise the transition temperature by an amount no volume ratio contains.

7.4 What the simulation assumes

The material budget is an idealized finished-specimen accounting: gross rectangular volume, integer specimen counts, remainder discarded. It is not a prediction of how many specimens fit in, or can be machined from, a real surveillance capsule; capsule layout, holders, orientation constraints, archive requirements and reconstitution are not modelled. Specimen scatter is independent conditional on the programme-level correction draw, so no within-programme correlated laboratory or instrument systematic is modelled — one machine, one operator, one calibration applied to every specimen. An independent error averages away with specimen count; a correlated one does not, so every large-count miniature gain reported here is conditional on that assumption and is potentially optimistic. No extra variance component was invented to cover it.

An absolute energy-resolution floor is named, quantified and deliberately not modelled: mapped back through its normalization factor, a typical material gives the smallest geometry a raw curve spanning well under a joule in total, smaller than anything in the dataset's own miniature fits. No citable resolution value was found in the sources read, so the effect is stated with its direction rather than fabricated.

Finally, one wording ambiguity in the adaptive placement algorithm was found after results existed. The pre-registered reading was kept, the alternative was not added, and the ambiguity is recorded here as a limitation rather than rewritten to look unambiguous. No full-size result depends on it.

7.5 Sensitivity to the inclusion criteria and the scatter model

Both were tested rather than asserted. Propagating the deliberately looser and deliberately stricter curve cohorts through the simulation changes no conclusion. Across every scatter-model variant — a robust-scale version, a smooth calibrated form, empirical resampling with and without its region means, and an inflated miniature profile — the median difference moves by about a degree at most and never changes sign. The inclusion rule and the scatter model are not what the answer rests on. The size correction is.

7.6 What the analysis could not determine

A geometry-resolved size correction. A product-form-specific correction for forging or weld metal. A correction specific to irradiated material, or whether irradiation and geometry interact. The separation of real material scatter from between-study scatter. A resolution floor for the smallest geometries. The true within-programme correlated error. And whether the choice of dataset release changes the downstream conclusions. This one needs stating precisely, because the project's own governing decision asked for it. The release comparison was performed through cleaning and curve definition, where the two releases differ by exactly two publications. It was not propagated through fitting, per-curve uncertainty, the size effect or the simulation. One of those two publications supplies one of the matched families behind the size correction, so the pooled correction values would move by an amount this work has not computed. The requirement to test the release choice downstream was therefore waived by an explicit author decision, on the ground that resolving it means re-running every downstream stage; it is not a requirement this report claims to have met. No claim is made that the choice of release has been shown irrelevant. The frozen correction sensitivities already span a wider range than this difference plausibly spans, and all of them move the answer the same way, which is a reason to expect the conclusion to survive — but that is a reasoned expectation, not a calculation performed on the earlier release.

7.7 Relationship to the 2026 machine-learning correlation paper

A recent study by the authors of this dataset uses machine learning to predict standard-size Charpy properties from sub-size measurements, on largely the same matched sub-population. That is a prediction problem; this is a design problem, and the two need different things from the same data.

The difference that matters is not accuracy but what the numbers mean. Their transformation builds each training target from the paired full-size curve, which is legitimate and necessary for supervised learning; their cross-validation holds out whole specimen sets, so the reported metric evaluates predictive performance on held-out specimen sets under that cross-validation split. It is not, and is not claimed by them to be, an independent estimate of how much uncertainty a classical correction adds to a test programme. This project needs precisely that second quantity, so its correction is never allowed to see the paired full-size curve — a restriction checked numerically on every run. Their benchmark of classical correlations is broader and more systematic than anything here.

The two sets of numbers are not comparable, and this report does not compare them. That study reports prediction and correlation performance against a full-size transition temperature, on its own matched subset, with its own index definition. This work reports the family-weighted bias and spread of the residual left by an out-of-sample classical correction applied to a different index, on a differently matched subset selected under different evidentiary filters. The metrics, the populations, the targets and the purposes all differ, so neither set of values validates, corroborates or bounds the other, and no magnitude comparison between them would mean anything. What can be compared is method: that study measures how accurately sub-size data predict full-size properties; this one measures how much uncertainty a classical correction contributes, and spends that uncertainty on a test-programme design question. One consequence is worth stating rather than leaving for a reader to notice: the project's own frozen size-effect summary in outputs/ does place this work's residual scale alongside the magnitudes that study reports for classical corrections. That summary is an approved generated output and is not edited retrospectively, but the comparison is not repeated here, for the reason just given.

No novelty is claimed beyond the following: a strictly matched, fully audited pairing in which every rejection reason is preserved; a correction applied out of sample; an explicit residual-uncertainty estimate carrying an evidence status; a statement of where the dataset is too thin to support one at all; and the propagation of that uncertainty into a test-matrix design answer. This project trains no model and attempts no prediction.

8. Conclusions

The main question has a quantitative answer with its conditions attached. Spending a fixed volume of irradiated material on many miniature specimens rather than few full-size ones improves the recovered full-size index temperature only while the material budget is very small. Under the central assumptions the miniature error reaches a floor of 11.96 °C — the root-sum-square of the analysis-derived correction bias and its residual standard deviation — and no number of additional miniature specimens lowers it, because the correction error is a property of the programme, not of the specimens. Miniature specimens win at the smallest budget tested for three of the four placement strategies, are placement-dependent and often tied around the next budget, win no comparison from a budget of 8 full-size equivalents, and from 18 lose every comparison. This is a conditional fixed-volume scenario using the pooled, WEAKLY SUPPORTED correction, under the tested assumptions; it is not a finding that full-size specimens are preferable in general.

The conditions are as much the result as the number is. The correction distribution rests on 7 independent families from 5 studies, no individual geometry is supported on its own, and the evidence is miniaturized rather than conventional sub-size. The correction assumption, not the scatter model, is what the answer depends on: the threshold correction uncertainty at which the comparison would turn is 5.3 to 9.7 °C at a budget of eight, against an analysis-derived 11.43 °C. And some comparisons cannot be made at all: for cold-transition materials the smaller geometries cannot physically reach their own transition.

For a test-programme designer, three practical conclusions follow. First, where the transition location is poorly known before testing, placement matters more than specimen size, and a clustered design that misses the transition produces no usable answer at any specimen count. Second, a confidence interval quoted on a miniature-derived full-size-equivalent index temperature describes sampling scatter only; it does not contain the size-correction error, and it becomes more misleading, not less, as specimens are added. Third, the most valuable thing that could be done for miniature-specimen surveillance is not more miniature specimens but better-supported correction data: matched full-size and miniature curves on the same heats, across product forms and irradiation states, which is exactly what this compilation is too thin to provide.

What would change the answer. A correction residual standard deviation below the strategy-dependent 5.3 to 9.7 °C thresholds at a budget of eight full-size equivalents, for the median true curve and geometry of this simulated scenario; a correction supported geometry by geometry rather than pooled; evidence that irradiation and geometry do not interact; or a demonstration that the correction error behaves partly like independent specimen noise rather than wholly like a programme-level offset. The last of these was tested as a labelled illustration and does reverse the conclusion, which is precisely why the programme-level treatment is argued for explicitly rather than assumed.

9. References

Sources are marked with what was actually read. Nothing is cited from memory.

- Lee, I., et al. Dataset of Charpy V-notch impact test records of nuclear structural materials. Materials Cloud Archive, records q4x6v-dyt25 and eny5v-www93. CC BY 4.0. Data files read in full.

- Lee, I., et al. Comprehensive Toughness Dataset of Nuclear Reactor Structural Materials using Charpy V-Notch Impact Testing. Scientific Data. Full text read, including its specimen-size-effect section and correlation tables.
- Sokolov, M. A., and Alexander, D. J. An Improved Correlation Procedure for Subsize and Full-Size Charpy Impact Specimen Data. NUREG/CR-6379. Full report read; source of the primary temperature correction and its stated applicability limits.
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- Lucon, E., Splett, J., Koepke, A., and Newton, D. NIST Software Package for Obtaining Charpy Transition Curves. NIST Technical Note 2158. Full text read; method reference.
- Kasala Sreenivasulu, Y., et al. Machine Learning-based Correlation of Charpy Impact Properties Between Sub-sized and Standard-sized Specimens for Nuclear Structural Materials. Scientific Reports; arXiv 2607.10412. Full text read; discussed in section 7.7.
- ASTM E23, E185, E2248, E1921, E900, ISO 148-1, and 10 CFR Part 50. Cited for specific clauses; each clause read is recorded with its use in the decision log.
- Lucas et al., Corwin et al., Louden et al., Kayano et al., Towers, and Wallin are cited second-hand, through the sources above that tabulate their equations; their original papers were not read, and every equation taken from them names the route.

Appendix A — Claim-to-source map

Every numeric claim in this report is substituted at build time from a table regenerated by the pipeline. The build fails if any placeholder is unresolved, if a row selector matches other than exactly one row, or if a literal result number appears in the prose. The complete map, one row per claim, giving the value, the generated file, the row selector and the column, is written to report/claim_source_map.csv alongside this document, and the derived quantities additionally record their inputs and the expression used. Every other literal number in this report — criterion energies, method constants, standard designations, cited coefficients — is a registered constant carrying a provenance category and a source, listed in report/constant_source_map.csv, so an externally cited value can be audited as readily as a generated one.

Appendix B — Reproducibility

One command. ./run_all.sh recreates the pinned environment if absent, refuses any other package versions, verifies the raw-data checksums, regenerates every output from the vendored raw data, and rebuilds this report. Its figures are the approved generated figures, embedded rather than redrawn, so the document cannot drift from the analysis.

Determinism and provenance. Every stochastic stage draws from an independent seeded stream per replicate, so results do not depend on worker count, chunking or execution order — tested, not asserted; figures and this PDF carry no document metadata, so repeated runs are byte-identical. The raw workbooks are never modified and their digests are verified on every run, and the curve cohorts frozen at the second checkpoint are re-checked by every later stage, which stops the pipeline on any drift. Each work package's generated summary carries the approval status it had when written, since changing it would mean rerunning an approved analysis; the decision log and progress file are authoritative for status.